Attention Is All You Need

The Transformer Architecture that Revolutionized NLP

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Google Brain & Google Research

Presented by-
Sajjan Singh

Presentation Outline

1. Introduction
2. The Transformer Architecture
3. Attention Mechanism
4. Position-wise Feed-Forward Networks
5. Embeddings and Positional Encoding
6. Why Self-Attention?
7. Training and Results
8. Conclusion and Impact

The Pre-Transformer Era

Dominant Sequence Models (Before 2017):

- **Recurrent Neural Networks (RNNs)**
  - Process sequences step-by-step
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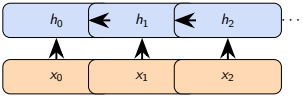
RNN Sequential Processing



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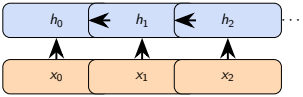

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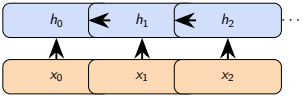

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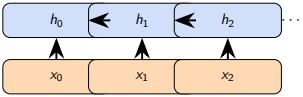

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Core Problem

Sequential computation prevents parallelization!

- Long sequences $\rightarrow$ slow training
- Memory constraints limit batching
- Difficulty learning long-range dependencies

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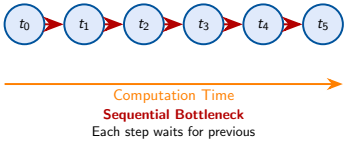

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Maximum path length between any two positions: $O(n)$

This makes learning **long-range dependencies** very difficult!

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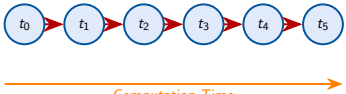
Goal: Process all positions simultaneously!



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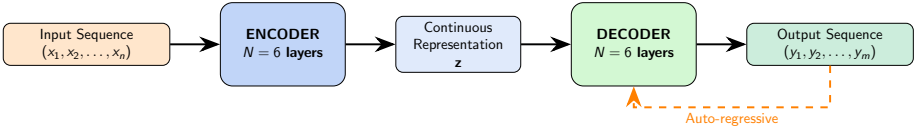
The Transformer Solution

Constant path length: $O(1)$

Uses **self-attention** to connect all positions directly!

The Big Picture: Encoder-Decoder Framework



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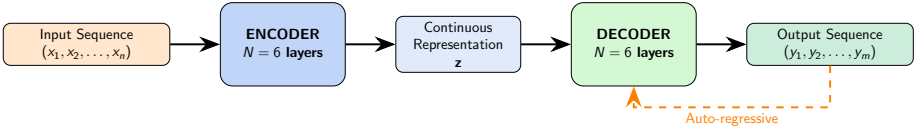


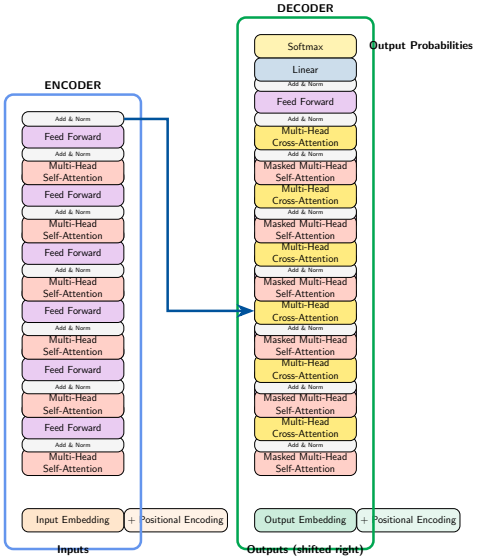
Encoder

Maps input $(x_1, \ldots, x_n)$ to continuous representations $\mathbf{z} = (z_1, \ldots, z_n)$

Decoder

Generates output $(y_1, \ldots, y_m)$ one element at a time, consuming previously generated symbols

Complete Transformer Architecture



Encoder Stack Details

Key Components:

1. **Multi-Head Self-Attention**
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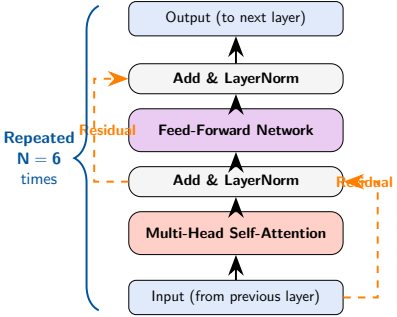

Dimension Requirements

All sub-layers produce outputs of dimension $d_{\text{model}} = 512$

This facilitates residual connections!

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Three Sub-layers per Layer:

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   - Prevents attending to future positions
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   - Masking: set to $-\infty$ before softmax

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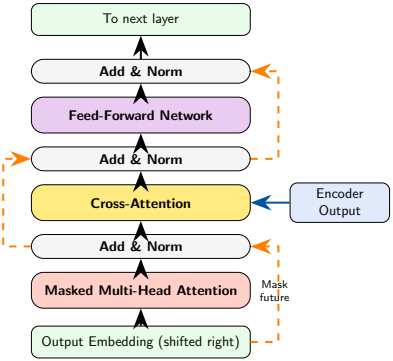

Critical: Masking

Predictions for position i can depend only on known outputs at positions less than i

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Definition

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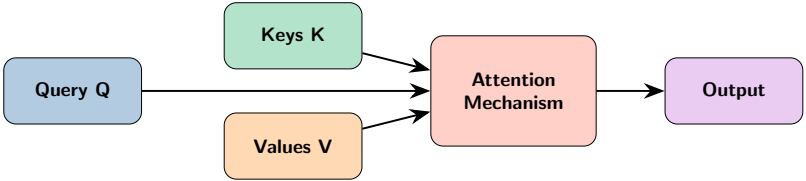

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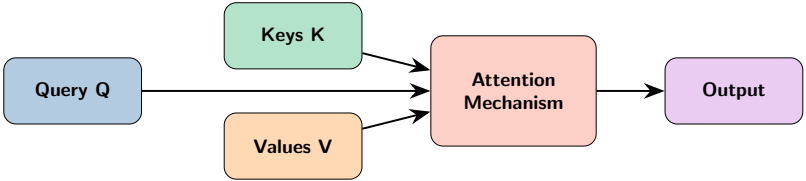

Output Computation:

- Weighted sum of values
- Weights from compatibility function
- Compatibility = query-key similarity

Intuition:

- Query: What I'm looking for
- Keys: What I can match against
- Values: What I retrieve

Scaled Dot-Product Attention

The Formula:

$$\boxed{\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V}$$

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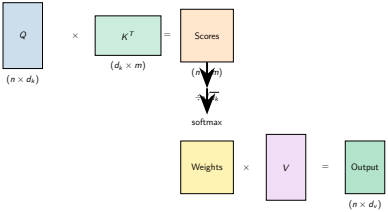

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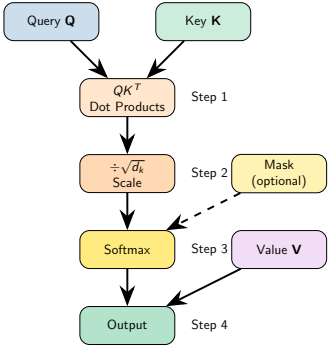
Why Scale by $\sqrt{d_k}$?

Large d_k makes dot products too large, so scaling keeps softmax gradients stable.

Scaled Dot-Product Attention: Step-by-Step



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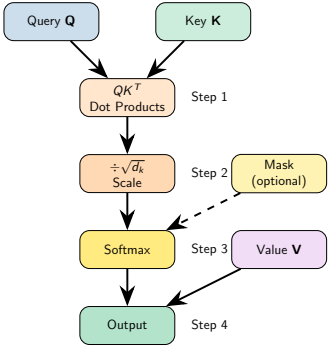

Quick Summary

$QK^T \to$ scale by $\sqrt{d_k} \to$ mask (decoder only) $\to$ softmax $\to$ weighted sum with V. Dot-product attention is faster and more memory-efficient than additive attention.

Multi-Head Attention: The Key Innovation

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Instead of single attention with
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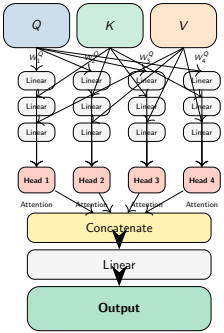

Formally:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O$$

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

Multi-Head Attention: Configuration

Paper's Configuration:

- Number of heads: $h = 8$
- Model dimension: $d_{\text{model}} = 512$
- Key dimension per head:
  $d_k = d_{\text{model}}/h = 64$
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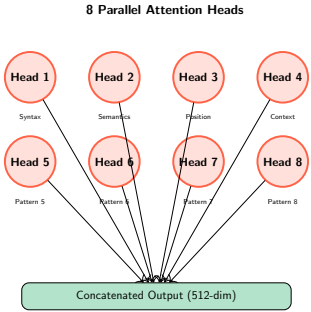

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Why Multiple Heads?

Allows the model to **jointly attend** to information from **different representation subspaces** at different positions.

Three Types of Attention in Transformer

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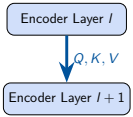

All positions attend
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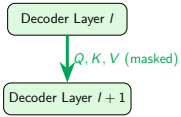
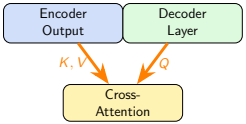
2. Masked Decoder Self-Attention



Each position attends
only to earlier positions

3. Encoder-Decoder Attention



Decoder attends to
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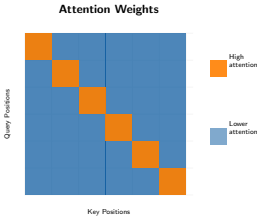

Interpretability Benefit

Self-attention yields more interpretable models. Individual attention heads clearly learn to perform different tasks, many exhibiting behavior related to syntactic and semantic structure.

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- ReLU activation in between
- Applied to **each position separately**
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Position-wise Feed-Forward Networks

Architecture:

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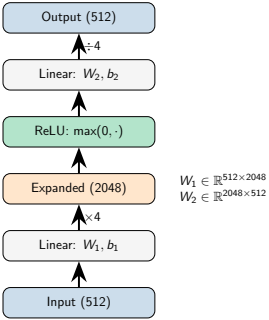

Dimensions

- Input/Output: $d_{\text{model}} = 512$
- Inner layer: $d_{\text{ff}} = 2048$

Alternative View: Convolutions

Equivalence to Convolutions

Another way of describing this is as **two convolutions with kernel size 1**.

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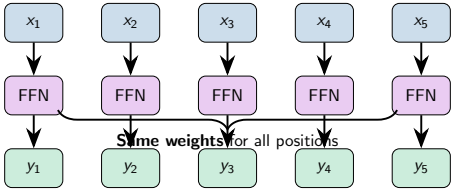
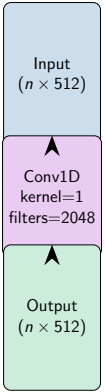
Position-wise FFN Application



Equivalent to Conv1D



Embeddings and Softmax

Token Embeddings:

- Convert input/output tokens to vectors
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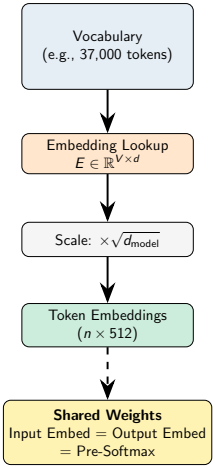
Why Scale Embeddings?

To make the embeddings and positional encodings comparable in magnitude when they are summed together.



Positional Encoding: The Key to Sequence Order

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Transformer has no recurrence and no convolution. How does it know the **order** of tokens in a sequence?

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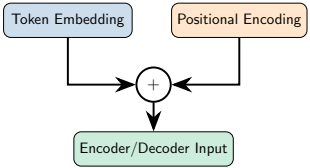

Sinusoidal Positional Encoding

The Encoding Functions:

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right)$$

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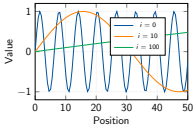

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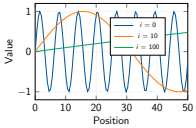

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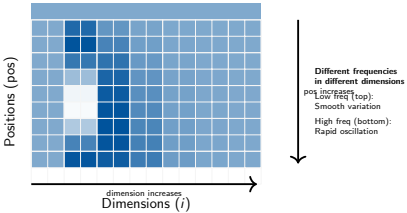
Key Property

For any fixed offset k, PE_{pos+k} can be represented as a **linear function** of PE_{pos}. This allows the model to easily learn to attend by relative positions!

Positional Encoding Visualization

Positional Encoding Matrix



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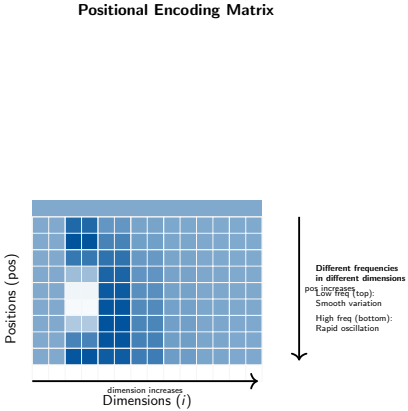

Extrapolation Benefit

Chose sinusoidal version because it may allow the model to **extrapolate to sequence lengths longer** than those encountered during training.

Comparison: Self-Attention vs. RNN vs. CNN

Layer Type	**Complexity per Layer**	**Sequential Ops**	**Max Path Length**
Self-Attention	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$
Convolutional	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k(n))$
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Where $n < d$

Self-attention is **faster** than recurrent layers when sequence length n is smaller than representation dimension d.

True for most state-of-the-art models!

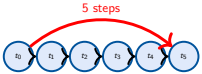
Key Advantage

Maximum path length between any two positions: $O(1)$

This is crucial for learning **long-range dependencies**!

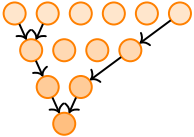
Visual Comparison: Path Lengths

RNN: $O(n)$



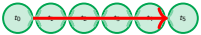
Long path for
distant positions

CNN: $O(\log n)$



Logarithmic path
(dilated convolutions)

Self-Attention: $O(1)$



Direct connection
in 1 step!

Three Desiderata for Sequence Models

1. Computational Complexity

Total computational cost per layer

Self-Attention: $O(n^2 \cdot d)$

2. Parallelization

Minimum sequential operations

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Self-Attention Wins on Two Fronts!

- **Constant path length** $\rightarrow$ easier to learn long-range dependencies
- **Constant sequential operations** $\rightarrow$ maximum parallelization
- Computational cost acceptable when $n < d$ (typical in practice)

Training Configuration

Datasets:

- **WMT 2014 English-German**
  - 4.5 million sentence pairs
  - Byte-pair encoding: 37,000 tokens
- **WMT 2014 English-French**
  - 36 million sentences
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- 8 NVIDIA P100 GPUs
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Learning Rate Schedule:

$$lrate = d_{\text{model}}^{-0.5} \cdot \min(\ldots)$$

- Warmup: 4,000 steps
- Linear increase then inverse sqrt decay

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Applied to:

- Output of each sub-layer
- Sums of embeddings and positional encodings

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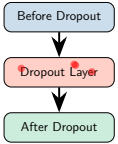

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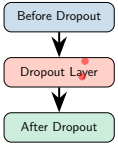

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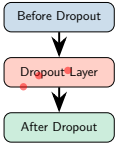

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Why Label Smoothing?

Makes the model learn to be more uncertain, which:

- Improves generalization
- Prevents overconfident predictions
- Better calibration

Results: Machine Translation Performance

Model	EN-DE BLEU	EN-FR BLEU	Training Cost
ByteNet [15]	23.75	–	–
Deep-Att + PosUnk [32]	–	39.2	1.0×10^{20}
GNMT + RL [31]	24.6	39.92	2.3×10^{19}
ConvS2S [8]	25.16	40.46	9.6×10^{18}
MoE [26]	26.03	40.56	2.0×10^{19}
GNMT + RL Ensemble	26.30	41.16	1.8×10^{20}
ConvS2S Ensemble	26.36	41.29	7.7×10^{19}
Transformer (base)	**27.3**	38.1	$\mathbf{3.3 \times 10^{18}}$
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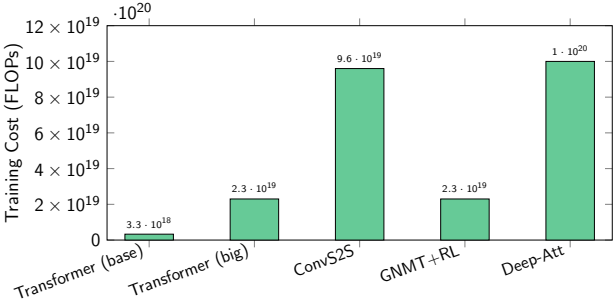
Takeaway

Transformer-big reaches **28.4** (EN-DE) and **41.0** (EN-FR) BLEU with much lower training cost than prior systems.

Training Cost Comparison



Key Insight

Transformer (base) achieves **better results** than all previous models at a **fraction of the training cost**!

Model Variations: Ablation Study

	Variation	PPL (dev)	BLEU (dev)	Params
	Base model	**4.92**	**25.8**	65M
(A)	$h = 1$	5.29	24.9	–
	$h = 4$	5.00	25.5	–
	$h = 16$	4.91	25.8	–
	$h = 32$	5.01	25.4	–
(B)	$d_k = d_v = 16$	5.16	25.1	58M
	$d_k = d_v = 32$	5.01	25.4	60M
(C)	$d_{model} = 1024$	4.66	26.0	168M
	$d_{ff} = 4096$	4.75	26.2	90M
(D)	Dropout $= 0.0$	5.77	24.6	–
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Takeaway

Best trade-off uses multi-head attention (not single-head), adequate key/value size, and dropout around 0.1.

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Attention Heads

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Learned vs. Sinusoidal: **Nearly identical results!**

Sinusoidal may allow longer sequence extrapolation.

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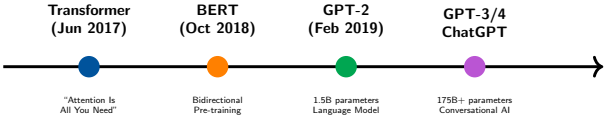
4. Interpretability

- Attention heads learn different patterns
- Clear syntactic/semantic behavior
- More interpretable than RNNs

The Transformer Revolution



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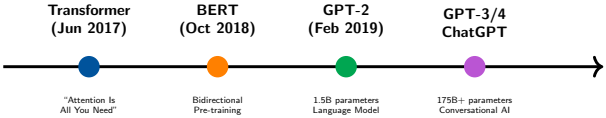

The Foundation of Modern NLP

The Transformer architecture has become the foundation for virtually all state-of-the-art NLP models:

- **BERT, RoBERTa, ALBERT** - Understanding tasks
- **GPT series** - Generation tasks
- **T5, BART** - Text-to-text frameworks
- **Vision Transformer (ViT)** - Computer vision
- **GPT-4, Claude, Gemini** - Large Language Models

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- Sparse attention patterns

Multi-Modal Extensions

Problems involving input/output modalities other than text:

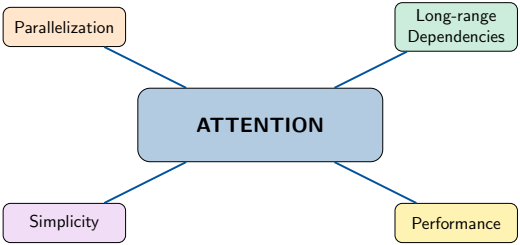
- Image captioning
- Visual question answering
- Speech-to-text

Making Generation Less Sequential

Research goal: parallel generation

- Non-autoregressive models
- Iterative refinement
- Speculative decoding

Summary: Why “Attention Is All You Need”



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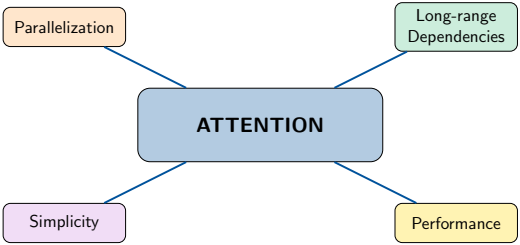

The Core Message

Attention mechanisms alone – without recurrence or convolution – are sufficient for high-quality sequence transduction.

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5. **Residual connections + LayerNorm** enable training deep networks
   - 6 layers in encoder and decoder
   - Stable gradient flow

References

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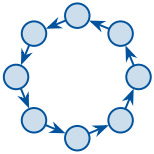
Thank You!

Questions?



Code: `https://github.com/tensorflow/tensor2tensor`